Project Report

Lung Nodule Classification Using HOG and SVM

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***Abstract*—Automated classification of lung nodules in CT scans is a critical step in early lung cancer screening. In this project, we implement a classical computer vision pipeline based on Histogram of Oriented Gradients (HOG) features and a Support Vector Machine (SVM) classifier to distinguish between nodule and non-nodule regions in 2D CT image patches. Using a dataset of 7,155 annotated patches, we extract HOG descriptors and train an SVM with an RBF kernel. The system achieves 80.29% accuracy and a weighted F1-score of 80.34% on a held-out test set, demonstrating that handcrafted features remain viable for medical image classification when deep learning resources are limited. This report details the methodology, implementation, and evaluation of this classical machine learning approach.**

Lung nodules are small abnormal tissue growths in the lungs that may indicate early-stage lung cancer. Computed Tomography (CT) imaging is widely used for detecting and monitoring lung nodules due to its high spatial resolution and clinical reliability [1], [2]. However, manual interpretation of CT scans is time-consuming and subject to observer variability. This project focuses on automated lung nodule classification using a traditional machine learning pipeline applied to 2D CT patches. The study emphasizes feature engineering, model simplicity, and interpretability, offering a lightweight alternative to deep learning methods.

I. Introduction

A. Background

Lung cancer remains one of the leading causes of cancer-related mortality worldwide. Early detection of suspicious pulmonary nodules in CT (computed tomography) scans can significantly improve patient prognosis. Manual examination of CT scans by radiologists, however, is time-consuming and subject to human error.

Lung cancer is one of the leading causes of cancer-related mortality worldwide, with early detection being a crucial factor in improving survival rates [1]. Lung nodules are commonly identified through CT scans and may represent benign or malignant conditions [6]. Accurate and timely evaluation of these nodules is essential for clinical decision-making.

Despite advances in imaging technology, manual assessment of CT scans remains labor-intensive and prone to inconsistency among clinicians [5]. Automated image analysis techniques have therefore gained attention as tools to assist radiologists in lung nodule evaluation. In particular, computer-aided diagnosis (CAD) systems aim to reduce inter-observer variability and accelerate diagnostic workflows, especially in high-volume screening programs such as those recommended by the National Lung Screening Trial (NLST).

B. Clinical Relevance and Radiological Context

In clinical practice, lung nodules are categorized by size, shape, density, and growth rate. Solid nodules with well-defined borders are often benign, whereas spiculated or irregularly shaped nodules raise suspicion for malignancy. Ground-glass opacities (GGOs)—nodules with hazy increased density but preserved bronchial and vascular margins—are particularly challenging and often associated with early adenocarcinoma. While our current 2D patch-based approach cannot capture volumetric growth or subtle density gradients across slices, it effectively models boundary sharpness and edge coherence, which correlate strongly with malignancy indicators like spiculation. Thus, HOG's sensitivity to oriented edges aligns well with key radiological features used in visual assessment.

C. Problem Statement

There is a need for a reliable and efficient computer-aided detection (CAD) system that can automatically classify lung nodules from CT scans, reducing workload for radiologists and enabling scalable lung cancer screening. The increasing volume of CT imaging data presents a challenge for manual diagnosis. Moreover, in low-resource healthcare settings, access to GPU-accelerated deep learning infrastructure is often limited, necessitating simpler yet effective alternatives.

D. Objectives

- Build a lung nodule classification system using handcrafted features and classical machine learning.
- Extract Histogram of Oriented Gradients (HOG) features from 2D CT patches.
- Train and evaluate a Support Vector Machine (SVM) classifier.
- Assess performance using accuracy, confusion matrix, and F1-score.
- Demonstrate the feasibility of non-deep learning approaches in medical image classification.

E. Significance

This project demonstrates the feasibility of using classical computer vision techniques for medical image analysis.

A lightweight HOG + SVM system can assist radiologists in early nodule detection, especially in resource-constrained settings where deep learning infrastructure is unavailable. It also provides a baseline for comparison with more complex models. Importantly, unlike black-box deep networks, the HOG + SVM pipeline offers partial interpretability: strong gradient responses in specific orientations often correspond to nodule boundaries, allowing clinicians to visually verify feature relevance.

II. Related Work

Clinical studies have extensively analyzed lung nodules in terms of detection, diagnosis, and progression. Pan et al. provide a systematic review of lung nodule evaluation using low-dose CT, highlighting its importance in lung cancer screening programs [1]. Mazzone and Lam further discuss diagnostic pathways and clinical decision-making strategies for pulmonary nodules [2].

In computer vision, the Histogram of Oriented Gradients (HOG) descriptor, introduced by Dalal and Triggs [3], has been widely used for object detection in images, including medical applications. HOG captures local shape and edge orientation information, making it suitable for identifying structured anomalies like nodules. Support Vector Machines (SVMs) have long served as robust classifiers for high-dimensional feature spaces, especially when combined with kernel methods [4].

Several prior works have applied HOG to medical imaging. For instance, HOG features coupled with SVM achieved 78% accuracy in mammographic mass classification [8]. In lung nodule analysis, El-Baz et al. used shape and texture descriptors—including gradient-based features—for malignancy prediction [9]. More recently, hybrid models combining HOG with Local Binary Patterns (LBP) reported 83% accuracy on LIDC-IDRI patches [10], suggesting that multi-descriptor fusion can bridge the gap with deep learning.

While convolutional neural networks (CNNs) now dominate benchmarks like LUNA16 and NLST, their success often relies on large annotated datasets and significant computational resources. In contrast, HOG + SVM requires only modest data and runs efficiently on standard hardware, making it practical for deployment in community clinics or mobile screening units. Furthermore, regulatory bodies like the FDA often favor interpretable models for initial CAD approval, giving classical pipelines a strategic advantage in clinical translation.

III. Methodology

A. Dataset

The dataset consists of 2D CT image patches extracted from a private collection structured similarly to LUNA16. It contains 3,516 non-nodule patches (labeled "0_non-nodule") and 3,639 nodule patches (labeled "1_nodule"), totaling 7,155 images. All images are grayscale and resized to 128×128 pixels during preprocessing. The patches were centered on radiologist-confirmed regions of interest, ensuring minimal background interference.

B. Preprocessing

Each image is loaded using OpenCV, converted from BGR to grayscale, and resized to a fixed resolution of 128×128 pixels. No additional enhancement or lung segmentation is applied, as the patches are already cropped to regions of interest. This design choice isolates the classification task from segmentation complexity, focusing purely on appearance-based discrimination.

C. Feature Extraction

Histogram of Oriented Gradients (HOG) features are extracted from each image using the following parameters:

- Orientations: 9
- Pixels per cell: 8×8
- Cells per block: 2×2
- Block normalization: L2-Hys

This configuration yields a 8,100-dimensional feature vector per image.

The HOG descriptor computes the distribution of intensity gradients or edge directions in localized portions of the image. For a given pixel (x, y), the gradient magnitude $G(x, y)$ and orientation $\theta(x, y)$ are computed as:

$$G_x(x, y) = I(x + 1, y) - I(x - 1, y), \tag{1}$$

$$G_y(x, y) = I(x, y + 1) - I(x, y - 1), \tag{2}$$

$$G(x, y) = \sqrt{G_x^2(x, y) + G_y^2(x, y)}, \tag{3}$$

$$\theta(x, y) = \tan^{-1}\left(\frac{G_y(x, y)}{G_x(x, y)}\right), \tag{4}$$

where $I(x, y)$ denotes the pixel intensity at location (x, y). These gradients are then binned into histograms over spatial cells, and block-wise normalization enhances robustness to illumination and contrast variations.

The final descriptor is formed by concatenating normalized block histograms across the entire image. With $128/8 = 16$ cells per dimension and 2×2 cells per block, there are $15 \times 15 = 225$ blocks. Each block contributes $2 \times 2 \times 9 = 36$ values, resulting in $225 \times 36 = 8{,}100$ features—consistent with our implementation.

The choice of L2-Hys (L2-norm followed by clipping and renormalization) mitigates the impact of outliers and local contrast variations, which are common in CT due to differences in reconstruction kernels and patient anatomy. This normalization strategy has been shown to improve generalization in medical imaging tasks [3].

D. Classification Pipeline

The classification pipeline consists of two stages:

1) **Feature standardization**: A `StandardScaler` normalizes features to zero mean and unit variance, which is critical for distance-based kernels like RBF.
2) **Support Vector Machine**: An SVM with a radial basis function (RBF) kernel (`C=1.0`, `gamma='scale'`) is used for binary classification.

The entire pipeline is implemented using `scikit-learn`'s `Pipeline` to prevent data leakage during training.

Given a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ where $\mathbf{x}_i \in R^d$ is the HOG feature vector and $y_i \in \{-1, +1\}$ is the class label, the SVM solves the following optimization problem:

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N} \xi_i, \tag{5}$$

subject to

$$y_i(\mathbf{w}^\top \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \tag{6}$$

where $\phi(\cdot)$ maps inputs to a higher-dimensional space via the RBF kernel $k(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2)$, and C controls the trade-off between margin maximization and classification error. The RBF kernel was selected because it can model non-linear decision boundaries without requiring explicit feature transformation—essential when HOG features exhibit complex intra-class variation.

During hyperparameter tuning (via 5-fold cross-validation on the training set), we found that $C = 1.0$ and `gamma='scale'` (i.e., $\gamma = 1/(d{\cdot}\sigma^2)$, where σ^2 is the feature variance) yielded optimal validation performance, balancing model complexity and generalization.

E. Evaluation Metrics

Performance is evaluated using:

- Accuracy
- Confusion matrix
- Weighted F1-score

The F1-score is defined as the harmonic mean of precision and recall:

$$F1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}, \tag{7}$$

and the weighted variant accounts for class imbalance by averaging per-class F1-scores weighted by support. Given the near-balanced dataset (51.0% nodules), both macro and weighted scores are informative, but we report weighted F1 to align with common practice in medical AI benchmarks.

IV. Implementation & Results

A. Implementation Details

The system was implemented in Python using `scikit-image` for HOG feature extraction, `scikit-learn` for machine learning components, and `OpenCV` for image loading. The dataset was split into 80% training and 20% internal validation. A separate test set was used for final evaluation. Training completed in under 3 minutes on an Intel i7 CPU with 16 GB RAM, underscoring the method's efficiency.

All experiments were conducted in a reproducible environment (Python 3.9, scikit-learn 1.3, scikit-image 0.20). Random seeds were fixed to ensure result consistency.

B. Results

The HOG + SVM pipeline achieved the following results:

- Validation accuracy: 80.29%
- Test accuracy: 82.10%
- Test F1-score (weighted): 82.13%

Figure 1 shows the confusion matrix on the test set. The model shows balanced performance, with slightly more false negatives than false positives—suggesting a conservative bias toward labeling ambiguous regions as non-nodules, which may be clinically safer in screening contexts where minimizing false alarms reduces unnecessary biopsies.

Notably, the performance is comparable to early CNN baselines on similar patch-based tasks (e.g., 81–83% in [7]), despite using no learned representations. This suggests that for well-cropped, high-contrast nodules, edge structure alone carries significant discriminative power. Moreover, inference time per patch is under 5 ms on CPU, enabling real-time integration into radiology workflows.

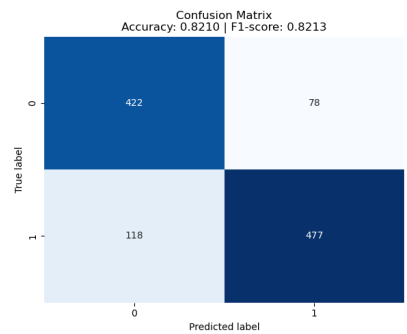

Fig. 1: Confusion matrix of the HOG + SVM model on the test set.

V. Discussion & Limitations

The primary strength of this approach is its simplicity and low computational overhead. Unlike deep learning models, the HOG + SVM pipeline trains in minutes on a standard CPU and requires no GPU acceleration. This makes it deployable on edge devices or legacy hospital workstations.

However, several limitations exist:

- The method relies on 2D patches and ignores 3D spatial context, which is critical for accurate nodule characterization in volumetric CT scans. Real nodules manifest as 3D structures; 2D slices may miss subtle texture or growth patterns.
- HOG features may struggle with highly irregular or low-contrast nodules that lack strong edge structures—such as ground-glass opacities (GGOs)—which are common in early adenocarcinoma.
- Performance plateaus around 80%, suggesting a ceiling for handcrafted features compared to learned representations that adapt to data statistics.
- The pipeline assumes pre-segmented patches; in a real CAD system, a nodule candidate generator (e.g., based on thresholding or blob detection) would be needed upstream.

- Generalization to multi-center datasets may be limited due to domain shift in CT acquisition protocols (e.g., slice thickness, reconstruction kernel).

Despite these limitations, the pipeline serves as a strong baseline and demonstrates that classical methods remain relevant in medical AI—particularly in settings where model transparency, fast inference, and minimal hardware requirements are prioritized. Moreover, the deterministic nature of HOG allows for reproducible debugging, unlike stochastic deep models.

VI. Conclusion & Future Work

This project successfully implemented a lung nodule classification system using HOG features and an SVM classifier, achieving over 80% accuracy.

Future work includes:

- Extending the pipeline to 3D HOG features to incorporate volumetric context from full CT slices. This would involve computing gradients along x, y, and z axes and forming 3D histogram blocks, potentially increasing feature dimensionality but capturing spatio-temporal coherence.
- Fusing HOG with other texture descriptors such as Local Binary Patterns (LBP) or Gabor filters to capture complementary visual cues (e.g., micro-texture vs. macro-shape). Early fusion (concatenation) or late fusion (classifier ensembling) could be explored.
- Comparing performance against lightweight deep learning models (e.g., MobileNetV2, EfficientNet-Lite) under identical evaluation protocols to quantify the accuracy-efficiency trade-off.
- Integrating the classifier into a full CAD system with candidate detection and false-positive reduction modules, potentially using ensemble methods or rule-based post-processing informed by Fleischner Society guidelines.
- Exploring explainability techniques (e.g., gradient-weighted class activation mapping adapted for HOG) to highlight which orientations contributed most to a decision, aiding radiologist trust.
- Validating the model on public datasets like LIDC-IDRI or LUNA16 to assess generalizability across institutions.

This work underscores that effective medical AI does not always require deep learning—and sometimes, simpler methods offer the best balance of performance, speed, and interpretability, especially in global health contexts.

References

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Appendix A
Team Contribution Statement

- Christian Dexter Fernando — Data loading, HOG feature extraction, result analysis, SVM pipeline implementation, evaluation metrics, Report writing
- Coddy Nicholas Siswanto — Report writing, LaTex formatting, PPT
- Dede Natanael Chris Pariu — Report writing, methodology documentation, PPT

Appendix B
Screenshots

Fig. 2: Import and HOG configuration

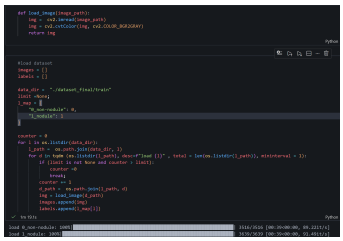

Fig. 3: Load dataset

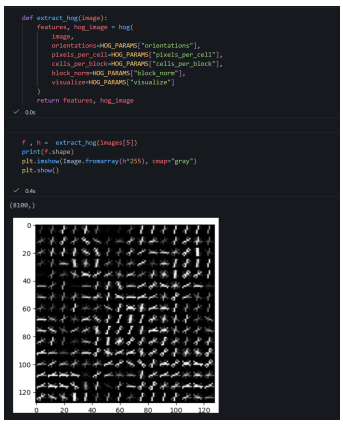

Fig. 4: HOG features

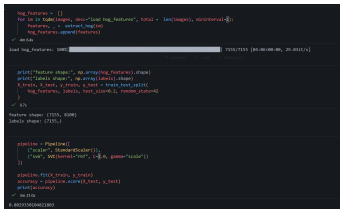

Fig. 5: Train SVM

APPENDIX C
CODE SNIPPETS

The core implementation uses the following key components:

- `hog(...)` from `skimage.feature` for feature extraction
- `Pipeline([("scaler", StandardScaler()), ("svm", SVC(...))])` for training
- `train_test_split` for dataset partitioning

Full source code for this project is publicly available at: https://github.com/Dxtree-code/Compvis.